High-Energy Expansion of Retarded Green's Function in Nambu Space

1 Setup

For a superconductor at equilibrium, the retarded Green's function in Nambu formalism is:

$$\mathbf{G}^R(\varepsilon) = [\varepsilon \mathbf{I} - \mathbf{H} - \mathbf{\Sigma}(\varepsilon)]^{-1} \tag{1}$$

The Hamiltonian in Nambu space is:

$$\mathbf{H} = \begin{pmatrix} \xi_{\mathbf{k}} & \Delta \\ \Delta^* & -\xi_{\mathbf{k}} \end{pmatrix} \tag{2}$$

where $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}} - \mu$ is the single-particle dispersion measured from the chemical potential, and Δ is the superconducting order parameter.

The self-energy matrix is:

$$\mathbf{\Sigma}(\varepsilon) = \begin{pmatrix} \Sigma_{11}(\varepsilon) & \Sigma_{12}(\varepsilon) \\ \Sigma_{21}(\varepsilon) & \Sigma_{22}(\varepsilon) \end{pmatrix} \tag{3}$$

1.1 Normalization Constraint

In the quasiclassical limit, the Green's function satisfies the fundamental normalization constraint:

$$\mathbf{g}^2 = \mathbf{I} \tag{4}$$

or equivalently:

$$(\mathbf{G}^R)^2 = -\mathbf{I} \tag{5}$$

This constraint reflects the fact that in Nambu space, the Green's function represents rotations in particle-hole space and must preserve the Nambu metric. For the 2×2 Nambu matrix:

$$\begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix}^2 = -\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{6}$$

This gives the component relations:

$$G_{11}^2 + G_{12}G_{21} = -1 \tag{7}$$
$$G_{11}G_{12} + G_{12}G_{22} = 0 \tag{8}$$
$$G_{21}G_{11} + G_{22}G_{21} = 0 \tag{9}$$
$$G_{21}G_{12} + G_{22}^2 = -1 \tag{10}$$

From the off-diagonal equations: $G_{12}(G_{11}+G_{22}) = 0$ and $G_{21}(G_{11}+G_{22}) = 0$.

2 High-ε Expansion

For large ε, we expand the Green's function using the geometric series. Define:

$$\mathbf{M} = \mathbf{H} + \mathbf{\Sigma} = \begin{pmatrix} \xi_{\mathbf{k}} + \Sigma_{11} & \Delta + \Sigma_{12} \\ \Delta^* + \Sigma_{21} & -\xi_{\mathbf{k}} + \Sigma_{22} \end{pmatrix} \tag{11}$$

Then:

$$\mathbf{G}^R(\varepsilon) = [\varepsilon\mathbf{I} - \mathbf{M}]^{-1} = \frac{1}{\varepsilon}\left[\mathbf{I} - \frac{\mathbf{M}}{\varepsilon}\right]^{-1} \tag{12}$$

Using the geometric series $(I - x)^{-1} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$:

$$\mathbf{G}^R(\varepsilon) = \frac{1}{\varepsilon}\sum_{n=0}^{\infty}\left(\frac{\mathbf{M}}{\varepsilon}\right)^n = \frac{1}{\varepsilon}\mathbf{I} + \frac{1}{\varepsilon^2}\mathbf{M} + \frac{1}{\varepsilon^3}\mathbf{M}^2 + \frac{1}{\varepsilon^4}\mathbf{M}^3 + \mathcal{O}(\varepsilon^{-5}) \tag{13}$$

3 Explicit Expansion Orders

3.1 Zeroth Order: $\mathcal{O}(\varepsilon^{-1})$

$$\mathbf{G}_0^R(\varepsilon) = \frac{1}{\varepsilon}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{14}$$

3.2 First Order: $\mathcal{O}(\varepsilon^{-2})$

$$\mathbf{G}_1^R(\varepsilon) = \frac{1}{\varepsilon^2}\begin{pmatrix} \xi_{\mathbf{k}} + \Sigma_{11} & \Delta + \Sigma_{12} \\ \Delta^* + \Sigma_{21} & -\xi_{\mathbf{k}} + \Sigma_{22} \end{pmatrix} \tag{15}$$

3.3 Second Order: $\mathcal{O}(\varepsilon^{-3})$

We need to compute $\mathbf{M}^2$:

$$\mathbf{M}^2 = \begin{pmatrix} \xi_{\mathbf{k}} + \Sigma_{11} & \Delta + \Sigma_{12} \\ \Delta^* + \Sigma_{21} & -\xi_{\mathbf{k}} + \Sigma_{22} \end{pmatrix}^2 \tag{16}$$

The matrix elements are:

$$\begin{aligned}
M_{11}^2 &= (\xi_{\mathbf{k}} + \Sigma_{11})^2 + (\Delta + \Sigma_{12})(\Delta^* + \Sigma_{21}) && (17)\\
M_{12}^2 &= (\xi_{\mathbf{k}} + \Sigma_{11})(\Delta + \Sigma_{12}) + (\Delta + \Sigma_{12})(-\xi_{\mathbf{k}} + \Sigma_{22}) && (18)\\
&= (\Delta + \Sigma_{12})(\Sigma_{11} + \Sigma_{22}) && (19)\\
M_{21}^2 &= (\Delta^* + \Sigma_{21})(\xi_{\mathbf{k}} + \Sigma_{11}) + (-\xi_{\mathbf{k}} + \Sigma_{22})(\Delta^* + \Sigma_{21}) && (20)\\
&= (\Delta^* + \Sigma_{21})(\Sigma_{11} + \Sigma_{22}) && (21)\\
M_{22}^2 &= (\Delta^* + \Sigma_{21})(\Delta + \Sigma_{12}) + (-\xi_{\mathbf{k}} + \Sigma_{22})^2 && (22)
\end{aligned}$$

Therefore:

$$\mathbf{G}_2^R(\varepsilon) = \frac{1}{\varepsilon^3}\begin{pmatrix} (\xi_{\mathbf{k}} + \Sigma_{11})^2 + |\Delta + \Sigma_{12}|^2 & (\Delta + \Sigma_{12})(\Sigma_{11} + \Sigma_{22}) \\ (\Delta^* + \Sigma_{21})(\Sigma_{11} + \Sigma_{22}) & |\Delta + \Sigma_{12}|^2 + (-\xi_{\mathbf{k}} + \Sigma_{22})^2 \end{pmatrix} \tag{23}$$

where we used $(\Delta + \Sigma_{12})(\Delta^* + \Sigma_{21}) = |\Delta + \Sigma_{12}|^2$ assuming $\Sigma_{21} = \Sigma_{12}^*$ (particle-hole symmetry).

3.4 Complete Expansion to $\mathcal{O}(\varepsilon^{-3})$

The full expansion without normalization constraint is:

$$\mathbf{G}^R(\varepsilon) = \frac{1}{\varepsilon}\mathbf{I} + \frac{1}{\varepsilon^2}\mathbf{M} + \frac{1}{\varepsilon^3}\mathbf{M}^2 + \mathcal{O}(\varepsilon^{-4}) \tag{24}$$

4 Expansion with Normalization Constraint $\mathbf{g}^2 = \mathbf{I}$

The normalization constraint $(\mathbf{G}^R)^2 = -\mathbf{I}$ fundamentally changes the expansion. We must find a parameterization that automatically satisfies this constraint.

4.1 Parameterization Satisfying Normalization

Any 2×2 matrix satisfying $\mathbf{g}^2 = \mathbf{I}$ can be parameterized as:

$$\mathbf{g} = \begin{pmatrix} g & f \\ f^* & -g \end{pmatrix} \tag{25}$$

where the constraint $g^2 + |f|^2 = 1$ must hold. This can be written as:

$$g = \cos\theta \tag{26}$$
$$f = e^{i\phi}\sin\theta \tag{27}$$

Alternatively, for the retarded Green's function with $(\mathbf{G}^R)^2 = -\mathbf{I}$:

$$\mathbf{G}^R = \begin{pmatrix} G & F \\ F^* & -G \end{pmatrix}, \quad G^2 + |F|^2 = -1 \tag{28}$$

This requires G to be purely imaginary at real frequencies. Writing $G = i\tilde{g}$ with $\tilde{g}$ real:

$$\boxed{\mathbf{G}^R = i\begin{pmatrix} \tilde{g} & -i\tilde{f} \\ -i\tilde{f}^* & -\tilde{g} \end{pmatrix}, \quad \tilde{g}^2 + |\tilde{f}|^2 = 1} \tag{29}$$

4.2 Constrained High-Energy Expansion

With the constraint, we parameterize:

$$\tilde{g}(\varepsilon) = \frac{1}{\sqrt{1 + |\tilde{f}(\varepsilon)|^2}} \tag{30}$$
$$\tilde{f}(\varepsilon) = \text{off-diagonal anomalous component} \tag{31}$$

For high ε, we expect $|\tilde{f}| \ll 1$, so we can expand:

$$\tilde{g}(\varepsilon) \approx 1 - \frac{1}{2}|\tilde{f}|^2 + \mathcal{O}(|\tilde{f}|^4) \tag{32}$$

From the Dyson equation $[\varepsilon\mathbf{I} - \mathbf{H} - \mathbf{\Sigma}]\mathbf{G}^R = \mathbf{I}$, we have:

$$(\varepsilon - \xi_{\mathbf{k}} - \Sigma_{11})iG - (\Delta + \Sigma_{12})F = 1 \tag{33}$$
$$-(\Delta^* + \Sigma_{21})F^* + (\varepsilon + \xi_{\mathbf{k}} - \Sigma_{22})iG = -1 \tag{34}$$

Using $G = i\tilde{g}$ and $F = -i\tilde{f}$:

$$-(\varepsilon - \xi_{\mathbf{k}} - \Sigma_{11})\tilde{g} - i(\Delta + \Sigma_{12})\tilde{f} = 1 \tag{35}$$
$$i(\Delta^* + \Sigma_{21})\tilde{f}^* - (\varepsilon + \xi_{\mathbf{k}} - \Sigma_{22})\tilde{g} = -1 \tag{36}$$

4.3 Leading Order with Constraint

At leading order in $1/\varepsilon$, taking $\tilde{g} \approx 1$ and $\tilde{f} \ll 1$:

$$-\varepsilon \approx 1 \quad \Rightarrow \quad \tilde{g}_0 = -\frac{1}{\varepsilon} \tag{37}$$

$$-i(\Delta + \Sigma_{12})\tilde{f} \approx 1 + \varepsilon\tilde{g}_0 = 0 \tag{38}$$

This shows that at leading order, $\tilde{f} \sim 1/\varepsilon$ as well. More precisely:

$$\mathbf{G}_0^R = -\frac{i}{\varepsilon}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{39}$$

But this violates $G^2 = -I$ unless we're more careful. The correct leading order satisfying the constraint is:

$$\boxed{\mathbf{G}^R(\varepsilon) = -\frac{i}{\sqrt{\varepsilon^2 - |\Delta + \Sigma_{12}|^2}}\begin{pmatrix} \varepsilon - \Sigma_{11} & \Delta + \Sigma_{12} \\ \Delta^* + \Sigma_{21} & -(\varepsilon - \Sigma_{22}) \end{pmatrix}} \tag{40}$$

where the normalization factor ensures $(\mathbf{G}^R)^2 = -\mathbf{I}$.

4.4 High-Energy Expansion with Normalization

At high energy, expand the normalization factor:

$$\frac{1}{\sqrt{\varepsilon^2 - |\Delta + \Sigma_{12}|^2}} = \frac{1}{\varepsilon}\frac{1}{\sqrt{1 - \frac{|\Delta+\Sigma_{12}|^2}{\varepsilon^2}}} \tag{41}$$

$$= \frac{1}{\varepsilon}\left(1 + \frac{|\Delta + \Sigma_{12}|^2}{2\varepsilon^2} + \mathcal{O}(\varepsilon^{-4})\right) \tag{42}$$

Therefore:

$$\boxed{\mathbf{G}^R(\varepsilon) = -\frac{i}{\varepsilon}\begin{pmatrix} 1 & \frac{\Delta+\Sigma_{12}}{\varepsilon} \\ \frac{\Delta^*+\Sigma_{21}}{\varepsilon} & -1 \end{pmatrix} + \mathcal{O}(\varepsilon^{-3})} \tag{43}$$

Including next order with Σ_{11} and Σ_{22}:

$$\boxed{\mathbf{G}^R(\varepsilon) = -\frac{i}{\varepsilon}\begin{pmatrix} 1 - \frac{\Sigma_{11}}{\varepsilon} & \frac{\Delta+\Sigma_{12}}{\varepsilon} \\ \frac{\Delta^*+\Sigma_{21}}{\varepsilon} & -1 + \frac{\Sigma_{22}}{\varepsilon} \end{pmatrix} + \mathcal{O}(\varepsilon^{-3})} \tag{44}$$

4.5 Comparison: With vs Without Normalization

Without constraint:

$$\mathbf{G}^R \approx \frac{1}{\varepsilon}\mathbf{I} + \frac{1}{\varepsilon^2}\mathbf{M} + \cdots \tag{45}$$

With constraint $(\mathbf{G}^R)^2 = -\mathbf{I}$:

$$\mathbf{G}^R \approx -\frac{i}{\varepsilon}\boldsymbol{\tau}_3 + \frac{1}{\varepsilon^2}\begin{pmatrix} -i\Sigma_{11} & -i(\Delta + \Sigma_{12}) \\ -i(\Delta^* + \Sigma_{21}) & i\Sigma_{22} \end{pmatrix} + \cdots \tag{46}$$

where $\boldsymbol{\tau}_3 = \mathrm{diag}(1, -1)$ is the third Pauli matrix.

The key differences:

- Factor of $-i$ appears in leading term
- Off-diagonal terms enter at $\mathcal{O}(\varepsilon^{-2})$ instead of $\mathcal{O}(\varepsilon^{-1})$
- Expansion automatically satisfies normalization at each order

5 Special Cases

5.1 Normal State ($\Delta = 0$, $\Sigma_{12} = \Sigma_{21} = 0$)

In the normal state with diagonal self-energy:

$$\mathbf{M} = \begin{pmatrix} \xi_{\mathbf{k}} + \Sigma_{11} & 0 \\ 0 & -\xi_{\mathbf{k}} + \Sigma_{22} \end{pmatrix} \tag{47}$$

The expansion becomes:

$$\mathbf{G}^R(\varepsilon) = \frac{1}{\varepsilon}\mathbf{I} + \frac{1}{\varepsilon^2}\begin{pmatrix} \xi_{\mathbf{k}} + \Sigma_{11} & 0 \\ 0 & -\xi_{\mathbf{k}} + \Sigma_{22} \end{pmatrix} + \frac{1}{\varepsilon^3}\begin{pmatrix} (\xi_{\mathbf{k}} + \Sigma_{11})^2 & 0 \\ 0 & (-\xi_{\mathbf{k}} + \Sigma_{22})^2 \end{pmatrix} + \cdots \tag{48}$$

5.2 Particle-Hole Symmetric Case ($\Sigma_{22} = -\Sigma_{11}$, $\mu = 0$)

With particle-hole symmetry, the off-diagonal elements simplify:

$$M^2_{12} = (\Delta + \Sigma_{12}) \cdot 0 = 0 \tag{49}$$
$$M^2_{21} = (\Delta^* + \Sigma_{21}) \cdot 0 = 0 \tag{50}$$

and the diagonal elements satisfy $M^2_{22} = M^2_{11}$.

6 Self-Energy Growing Faster than Linear

6.1 General Scaling Analysis

Consider the case where the self-energy grows faster than the linear ε term at high energy. Assume the asymptotic behavior:

$$\Sigma_{ij}(\varepsilon) \sim A_{ij}\varepsilon^{\alpha_{ij}} \quad \text{as } \varepsilon \to \infty \tag{51}$$

where $\alpha_{ij} > 1$ for some components.

Let $\alpha_{\max} = \max_{ij}\{\alpha_{ij}\}$ be the largest power. The Green's function is:

$$\mathbf{G}^R(\varepsilon) = [\varepsilon\mathbf{I} - \mathbf{H} - \mathbf{\Sigma}(\varepsilon)]^{-1} \tag{52}$$

Key observation: When $\alpha_{\max} > 1$, the self-energy dominates over the bare ε term at high energy. We must reorganize the expansion by factoring out the dominant contribution.

6.2 Reorganized Expansion

Define the dominant self-energy matrix $\mathbf{\Sigma}_{\text{dom}}(\varepsilon)$ containing only the terms with $\alpha_{ij} = \alpha_{\max}$, and write:

$$\mathbf{\Sigma}(\varepsilon) = \mathbf{\Sigma}_{\text{dom}}(\varepsilon) + \mathbf{\Sigma}_{\text{sub}}(\varepsilon) \tag{53}$$

where $\mathbf{\Sigma}_{\text{sub}}$ contains subdominant terms.

The Green's function becomes:

$$\mathbf{G}^R(\varepsilon) = [-\mathbf{\Sigma}_{\text{dom}}(\varepsilon) - (\varepsilon\mathbf{I} - \mathbf{H} - \mathbf{\Sigma}_{\text{sub}}(\varepsilon))]^{-1} \tag{54}$$

Factor out $\mathbf{\Sigma}_{\text{dom}}$:

$$\mathbf{G}^R(\varepsilon) = -\mathbf{\Sigma}_{\text{dom}}^{-1}(\varepsilon)\left[\mathbf{I} + \mathbf{\Sigma}_{\text{dom}}^{-1}(\varepsilon)\left(\varepsilon\mathbf{I} - \mathbf{H} - \mathbf{\Sigma}_{\text{sub}}(\varepsilon)\right)\right]^{-1} \tag{55}$$

Define the small parameter:

$$\mathbf{N}(\varepsilon) = \mathbf{\Sigma}_{\text{dom}}^{-1}(\varepsilon)\left(\varepsilon\mathbf{I} - \mathbf{H} - \mathbf{\Sigma}_{\text{sub}}(\varepsilon)\right) \tag{56}$$

Since $\mathbf{\Sigma}_{\text{dom}} \sim \varepsilon^{\alpha_{\max}}$, we have $\mathbf{N} \sim \varepsilon^{1-\alpha_{\max}} \to 0$ as $\varepsilon \to \infty$ when $\alpha_{\max} > 1$. Using the geometric series:

$$\mathbf{G}^R(\varepsilon) = -\mathbf{\Sigma}_{\text{dom}}^{-1}(\varepsilon)\sum_{n=0}^{\infty}(-1)^n\mathbf{N}^n(\varepsilon) \tag{57}$$

6.3 Explicit Example: Diagonal Dominant Self-Energy

Consider the case where only the diagonal elements dominate:

$$\mathbf{\Sigma}_{\mathrm{dom}}(\varepsilon) = \begin{pmatrix} A_{11}\varepsilon^{\alpha} & 0 \\ 0 & A_{22}\varepsilon^{\alpha} \end{pmatrix}, \quad \alpha > 1 \tag{58}$$

and the subdominant part includes off-diagonal anomalous terms:

$$\mathbf{\Sigma}_{\mathrm{sub}}(\varepsilon) = \begin{pmatrix} B_{11}\varepsilon^{\beta_{11}} & A_{12}\varepsilon^{\beta_{12}} \\ A_{21}\varepsilon^{\beta_{21}} & B_{22}\varepsilon^{\beta_{22}} \end{pmatrix} \tag{59}$$

where $\beta_{ij} < \alpha$.

Then:

$$\mathbf{\Sigma}_{\mathrm{dom}}^{-1}(\varepsilon) = \begin{pmatrix} \frac{1}{A_{11}\varepsilon^{\alpha}} & 0 \\ 0 & \frac{1}{A_{22}\varepsilon^{\alpha}} \end{pmatrix} \tag{60}$$

The expansion parameter is:

$$\mathbf{N}(\varepsilon) = \begin{pmatrix} \frac{1}{A_{11}\varepsilon^{\alpha}} & 0 \\ 0 & \frac{1}{A_{22}\varepsilon^{\alpha}} \end{pmatrix} \begin{pmatrix} \varepsilon + \xi_{\mathbf{k}} - B_{11}\varepsilon^{\beta_{11}} & \Delta + A_{12}\varepsilon^{\beta_{12}} \\ \Delta^* + A_{21}\varepsilon^{\beta_{21}} & \varepsilon - \xi_{\mathbf{k}} - B_{22}\varepsilon^{\beta_{22}} \end{pmatrix} \tag{61}$$

$$= \begin{pmatrix} \frac{\varepsilon}{A_{11}\varepsilon^{\alpha}} + \mathcal{O}(\varepsilon^{-\alpha}) & \frac{\Delta + A_{12}\varepsilon^{\beta_{12}}}{A_{11}\varepsilon^{\alpha}} \\ \frac{\Delta^* + A_{21}\varepsilon^{\beta_{21}}}{A_{22}\varepsilon^{\alpha}} & \frac{\varepsilon}{A_{22}\varepsilon^{\alpha}} + \mathcal{O}(\varepsilon^{-\alpha}) \end{pmatrix} \tag{62}$$

6.4 Leading Order Terms

The zeroth order (leading term) is:

$$\boxed{\mathbf{G}_0^R(\varepsilon) = -\begin{pmatrix} \frac{1}{A_{11}\varepsilon^{\alpha}} & 0 \\ 0 & \frac{1}{A_{22}\varepsilon^{\alpha}} \end{pmatrix} \sim \varepsilon^{-\alpha}} \tag{63}$$

The first correction is:

$$\mathbf{G}_1^R(\varepsilon) = \mathbf{\Sigma}_{\mathrm{dom}}^{-1}(\varepsilon)\mathbf{N}(\varepsilon) \tag{64}$$

Explicitly:

$$\mathbf{G}_1^R(\varepsilon) = \begin{pmatrix} \frac{1}{A_{11}\varepsilon^{\alpha}} & 0 \\ 0 & \frac{1}{A_{22}\varepsilon^{\alpha}} \end{pmatrix} \begin{pmatrix} \frac{\varepsilon}{A_{11}\varepsilon^{\alpha}} + \cdots & \frac{\Delta}{A_{11}\varepsilon^{\alpha}} + \cdots \\ \frac{\Delta^*}{A_{22}\varepsilon^{\alpha}} + \cdots & \frac{\varepsilon}{A_{22}\varepsilon^{\alpha}} + \cdots \end{pmatrix} \tag{65}$$

$$= \begin{pmatrix} \frac{1}{A_{11}^2\varepsilon^{2\alpha-1}} + \mathcal{O}(\varepsilon^{-2\alpha}) & \frac{\Delta}{A_{11}^2\varepsilon^{2\alpha}} + \mathcal{O}(\varepsilon^{\beta_{12}-2\alpha}) \\ \frac{\Delta^*}{A_{22}^2\varepsilon^{2\alpha}} + \mathcal{O}(\varepsilon^{\beta_{21}-2\alpha}) & \frac{1}{A_{22}^2\varepsilon^{2\alpha-1}} + \mathcal{O}(\varepsilon^{-2\alpha}) \end{pmatrix} \tag{66}$$

6.5 Power Counting Summary

For self-energy with dominant scaling $\Sigma \sim \varepsilon^\alpha$ where $\alpha > 1$:

- **Leading order:** $\mathbf{G}^R \sim -\mathbf{\Sigma}_{\text{dom}}^{-1} \sim \varepsilon^{-\alpha}$
- **First correction:** $\mathbf{G}_1^R \sim \varepsilon^{1-2\alpha}$ (diagonal) and $\sim \varepsilon^{-2\alpha}$ (off-diagonal with finite Δ)
- ***n*-th correction:** $\mathbf{G}_n^R \sim \varepsilon^{n(1-\alpha)}$ to $\varepsilon^{-n\alpha}$ depending on matrix elements

The expansion converges faster than the standard case because $|\mathbf{N}| \sim \varepsilon^{1-\alpha} \ll 1$ for $\alpha > 1$.

6.6 Physical Interpretation

When $\alpha > 1$, the self-energy represents **superlinear scattering** at high energies, which can arise from:

- Strong-coupling regimes beyond quasi-particle approximation
- Non-Fermi liquid behavior (e.g., $\alpha = 3/2$ in certain quantum critical systems)
- High-frequency cutoff effects in effective theories
- Resonant scattering off collective modes

In this regime, the system is **self-energy dominated** rather than kinetic-energy dominated, and the Green's function decays faster than $1/\varepsilon$ at high energies.

6.7 Normalization Constraint with $\Sigma \sim \varepsilon^\alpha$, $\alpha > 1$

When the self-energy grows superlinearly, the normalization constraint $(\mathbf{G}^R)^2 = -\mathbf{I}$ requires:

$$\mathbf{G}^R(\varepsilon) = -i \frac{\boldsymbol{\tau}_3 + \delta\mathbf{g}(\varepsilon)}{\sqrt{\text{Det}[\varepsilon\mathbf{I} - \mathbf{H} - \mathbf{\Sigma}]}} \tag{67}$$

For diagonal dominant self-energy $\Sigma_{ii} \sim A_{ii}\varepsilon^{\alpha}$, the determinant is:

$$\begin{aligned}
\text{Det} &= (\varepsilon - \xi_{\mathbf{k}} - \Sigma_{11})(\varepsilon + \xi_{\mathbf{k}} - \Sigma_{22}) - |\Delta + \Sigma_{12}|^2 && (68)\\
&\approx (\varepsilon - A_{11}\varepsilon^{\alpha})(\varepsilon - A_{22}\varepsilon^{\alpha}) - |\Delta|^2 && (69)\\
&\approx \varepsilon^{2\alpha} A_{11} A_{22} \left(1 - \frac{\varepsilon^{1-\alpha}}{A_{11}} - \frac{\varepsilon^{1-\alpha}}{A_{22}} + \cdots\right) && (70)
\end{aligned}$$

Taking the square root:

$$\sqrt{\text{Det}} \approx \varepsilon^{\alpha}\sqrt{A_{11}A_{22}} \left(1 - \frac{\varepsilon^{1-\alpha}}{2A_{11}} - \frac{\varepsilon^{1-\alpha}}{2A_{22}} + \cdots\right) \tag{71}$$

Therefore, the normalized Green's function at leading order is:

$$\boxed{\mathbf{G}^R(\varepsilon) \approx -\frac{i}{\varepsilon^{\alpha}\sqrt{A_{11}A_{22}}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \mathcal{O}(\varepsilon^{1-2\alpha})} \tag{72}$$

The decay is $\sim \varepsilon^{-\alpha}$ instead of $\sim \varepsilon^{-1}$, consistent with the unconstrained case but now properly normalized.

The first correction includes both the off-diagonal terms and the $\varepsilon^{1-\alpha}$ corrections:

$$\mathbf{G}_1^R(\varepsilon) = -\frac{i}{\varepsilon^{\alpha}\sqrt{A_{11}A_{22}}} \begin{pmatrix} \frac{\varepsilon^{1-\alpha}}{2A_{11}} & \frac{\Delta+\Sigma_{12}}{\varepsilon^{\alpha}\sqrt{A_{11}A_{22}}} \\ \frac{\Delta^*+\Sigma_{21}}{\varepsilon^{\alpha}\sqrt{A_{11}A_{22}}} & -\frac{\varepsilon^{1-\alpha}}{2A_{22}} \end{pmatrix} \tag{73}$$

7 General n-th Order

7.1 Standard Case (Σ finite or $\alpha \leq 1$)

The general expansion is:

$$\mathbf{G}^R(\varepsilon) = \sum_{n=0}^{\infty} \frac{1}{\varepsilon^{n+1}} \mathbf{M}^n \tag{74}$$

Each power of $\mathbf{M}$ contains increasingly higher-order products of the bare Hamiltonian parameters and self-energy components.

7.2 Superlinear Case ($\alpha > 1$)

The general expansion is:

$$\mathbf{G}^R(\varepsilon) = -\sum_{n=0}^{\infty}(-1)^n \mathbf{\Sigma}_{\text{dom}}^{-1}(\varepsilon)\mathbf{N}^n(\varepsilon) \tag{75}$$

where the expansion parameter $\mathbf{N} \sim \varepsilon^{1-\alpha}$ ensures rapid convergence.

8 Summary: Complete High-Energy Expansions

8.1 Case 1: Finite Σ, No Normalization Constraint

$$\mathbf{G}^R(\varepsilon) = \frac{1}{\varepsilon}\mathbf{I} + \frac{1}{\varepsilon^2}\begin{pmatrix} \xi_{\mathbf{k}} + \Sigma_{11} & \Delta + \Sigma_{12} \\ \Delta^* + \Sigma_{21} & -\xi_{\mathbf{k}} + \Sigma_{22} \end{pmatrix} + \mathcal{O}(\varepsilon^{-3}) \tag{76}$$

8.2 Case 2: Finite Σ, With Normalization $(\mathbf{G}^R)^2 = -\mathbf{I}$

$$\mathbf{G}^R(\varepsilon) = -\frac{i}{\varepsilon}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \frac{1}{\varepsilon^2}\begin{pmatrix} -i\Sigma_{11} & -i(\Delta + \Sigma_{12}) \\ -i(\Delta^* + \Sigma_{21}) & i\Sigma_{22} \end{pmatrix} + \mathcal{O}(\varepsilon^{-3}) \tag{77}$$

8.3 Case 3: $\Sigma \sim \varepsilon^\alpha$ ($\alpha > 1$), No Normalization

$$\mathbf{G}^R(\varepsilon) = -\begin{pmatrix} \frac{1}{A_{11}\varepsilon^\alpha} & 0 \\ 0 & \frac{1}{A_{22}\varepsilon^\alpha} \end{pmatrix} + \frac{1}{\varepsilon^{2\alpha-1}}\begin{pmatrix} \frac{1}{A_{11}^2} & 0 \\ 0 & \frac{1}{A_{22}^2} \end{pmatrix} + \mathcal{O}(\varepsilon^{-2\alpha}) \tag{78}$$

8.4 Case 4: $\Sigma \sim \varepsilon^\alpha$ ($\alpha > 1$), With Normalization

$$\mathbf{G}^R(\varepsilon) = -\frac{i}{\varepsilon^\alpha\sqrt{A_{11}A_{22}}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \mathcal{O}(\varepsilon^{1-2\alpha}) \tag{79}$$

8.5 Key Observations

1. **Normalization changes the prefactor:** Without normalization, $\mathbf{G}^R \sim 1/\varepsilon$; with normalization, $\mathbf{G}^R \sim -i/\varepsilon$ with Pauli matrix structure.

2. **Off-diagonal terms:**
   - Without normalization: appear at $\mathcal{O}(\varepsilon^{-1})$
   - With normalization: suppressed to $\mathcal{O}(\varepsilon^{-2})$
3. **Superlinear self-energy dominates:** When $\Sigma \sim \varepsilon^{\alpha}$ with $\alpha > 1$, the leading behavior is $\mathbf{G}^R \sim \varepsilon^{-\alpha}$, faster decay than standard $1/\varepsilon$.
4. **Normalization becomes exact in certain limits:**
   - Quasiclassical limit (integrate over $\xi_{\mathbf{k}}$)
   - Dirty limit (strong impurity scattering)
   - Usadel/Eilenberger equations
5. **Self-energy structure matters:**
   - Diagonal Σ: particle and hole sectors decouple
   - Off-diagonal Σ_{12}, Σ_{21}: anomalous self-energy, crucial for superconductivity
   - Particle-hole symmetry: $\Sigma_{22} = -\Sigma_{11}$ simplifies expansions